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Spatial Analysis of Criminality Based on Open Data of Czech Police (PČR)

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Abstract

This thesis aims to examine the effects of various socioeconomic factors on the prevalence of drug-related criminal offenses in the Czech Republic and to uncover the most problematic geographical areas. The methodology is based on the application of exploratory spatial data analysis techniques and statistical modeling, including the use of spatial econometric models and models for count data. The analysis reveals three clusters of administrative units exhibiting elevated crime rates, all located in the northern or north-eastern regions of Czechia. The results of the regression analysis based on the negative binomial model indicate that the divorce rate, and the proportion of young men and individuals with low education in the population significantly increase drug-related crime rates. The positive effect of the unemployment rate is not found to be statistically significant. A change in the share of foreigners and the number of living allowances per 1 000 inhabitants is associated with a decrease in the crime rate; however, both of these effects are likewise shown to be insignificant.

Keywords

drug-related criminality, negative binomial model, regression analysis, spatial econometrics

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Introduction

The Czech Republic occupies a specific position on the European drug market, particularly in the production and availability of methamphetamine and cannabis. Recent analyses of wastewater samples from 128 European cities, conducted by the European Union Drug Agency (2025), have highlighted Czechia as one of the leading countries in methamphetamine residues found in municipal wastewater. According to the National Drugs and Addiction Monitoring Center (2024), 189 methamphetamine labs were discovered in the Czech Republic in 2023. Smaller domestic production facilities predominate, supplying mainly the local consumer markets. Meanwhile, large-scale operations are increasingly being relocated abroad, particularly to Poland, Germany, the Netherlands, or Belgium. The main precursor used in methamphetamine production—pseudoephedrine—is primarily imported from Poland, further illustrating the transnational nature of drug trade. The prevalence of drug use in Czechia is alarming and contributes not only to public health concerns but also to a spectrum of criminal activities. While the overall crime rate in the Czech Republic has declined in recent years, the proportion of primary drug-related offenses has remained stable, accounting for approximately 2 % – 3 % of total recorded criminal activity (Scheinost et al., 2022).

This thesis aims to examine the impact of various socioeconomic factors on drug-related criminal activities in pursuit of understanding the motives behind the involvement in the drug trade. The main goal lies in identifying significant variables and quantifying their effect on crime rates. An additional objective is to study the spatial patterns of these offenses to assess whether high-crime areas are geographically clustered and identify the most problematic regions exhibiting high concentrations of crime.

The first two chapters are dedicated to literature review. Chapter 1 introduces the reader to the seminal works regarding the economics of crime and the elementary principles of sociology that explain how an individual's place in the societal structures and the established cultural norms influence the occurrence of crime. Moreover, it outlines the basic concepts of criminology theories and reasons the importance of empirical research in this field. Chapter 2 describes the methodological framework for spatial data analysis, including techniques such as Moran's *I* statistic and local indicators of spatial association. It explores key concepts in spatial econometric modeling and concludes with a discussion on the application of models for count data. The empirical analysis is presented in Chapter 3. The methodology includes data description along with manipulation process carried out in the *R Statistical Software* (v4.4.2; R Core Team, 2024), mainly with the use of packages such as *RCzechia* (Lacko, 2023), *spdep*, and *spatialreg* (Pebesma & Bivand, 2023). The results of regression analyses include the estimates of the classical linear regression model, four specifications of spatial models, and Poisson and negative binomial models. The key findings are discussed in Chapter 4.

1. Crime in a Socio-Economic Context

1.1 The Cost of Crime

One of the first most influential works in the field of economics of crime is Gary S. Becker's article from 1968, "*Crime and Punishment: An Economic Approach*". Before Becker, crime was only studied from the standpoint of other academic disciplines, mainly law, sociology, and psychology, which put emphasis on biological and sociological factors, such as biological inheritance or the environment in which an individual is raised. As opposed to them, Becker strictly focused on examining the economic benefits related to committing criminal activity. The motivation behind his research is derived from traditional economic theory of rational choice, in which an individual acts in such a way that maximizes his utility. A criminal is "born", when a person's expected utility from committing an offense surpasses his utility from engaging in other (legal) activities. This expected utility can be expressed mathematically as

$$EU_j = p_j U_j(Y_j - f_j) + (1 - p_j) U_j(Y_j), \quad (1.1)$$

where U_j is an individual's utility function, Y_j is his income from offense, f_j is the monetary equivalent of punishment and p_j is the probability of conviction (Becker, 1968). Therefore, the number of crimes committed by an individual is determined by

$$O_j = O_j(p_j, f_j, u_j), \quad (1.2)$$

where u_j is a variable covering all other influences (among other things, it depends on p_j and f_j for other offenses, because they can be seen as substitutes). The general consensus is that

$$O_{p_j} = \frac{\partial O_j}{\partial p_j} < 0, \quad (1.3)$$

$$O_{f_j} = \frac{\partial O_j}{\partial f_j} < 0, \quad (1.4)$$

because an increase in either the probability of conviction or the punishment leads to decreased expected utility and tendency to limit the number of committed crimes.

The focal point of Becker's work is to draw attention to the societal context of criminal behavior. It lies in formulating a function that defines the total social loss from offenses, that is

$$L = D(O) + C(p, O) + bpfO, \quad (1.5)$$

where $D(O)$ is the net cost or damage to society (the difference between the harm and gain to offenders), $C(O)$ are the costs of police and court activities and $bpfO$ is the total social loss from punishments (pO is the number of punished offenses and bf accounts for the loss per punished offense, b is a coefficient that transforms offender costs into social costs, which varies according to the type of punishment – close to zero for fines, higher than one for imprisonment). The purpose of the model is to determine the optimal values of p and f , which would minimize the total social loss.

Whether criminals are more easily deterred by a higher probability of conviction or punishment depends on their nature. If an individual is averse to risk, his expected utility can be reduced more significantly by higher punishment; therefore it is efficient to raise f and reduce p , making punishment severe but rare. However, if an individual has preference for risk, he is more deterred by higher probability of conviction. If it is believed that criminals are rather risk-seeking individuals, the effective judicial system should be built upon the idea of ensuring a higher probability of conviction (by allocating financial resources to police, courts, etc.). In Becker's economic perspective, however, we can sense a slight preference for increasing punishment severity (f) over probability of apprehension, due to the simple fact that the latter comes with higher enforcement cost that may not always be efficient.

Becker's economic theory of crime is to some extent criticized by various authors for several reasons. Polinsky and Shavell (1991) propose an idea that, given the realistic assumption that the wealth of individuals varies, non-maximal fines may be more desirable and the optimal fine should be such that only low-wealth individuals pay everything they have, as opposed to Becker's original idea of maximizing the fine. Skogh (1973), among other things, draws attention to one crucial component of Becker's model, that the net harm to society (D) is assumed to be greater than zero. However, there is no guarantee to that, quite the opposite, since the high motivation behind committing an offense roots from the vision of very "high" profit. In addition, property offenses can be seen as a form of transfer, suggesting the equality of harm to victims and net gain to offenders.

The fundamental concept of the model begins to break down when we start to consider crimes that do not have a direct victim (drug possession, traffic violations, prostitution, etc.). In such cases, it is somewhat challenging to quantify their social harm. One possibility is to view the consequences of such offenses as negative externalities, but the question remains as to how valid this approach is. Why don't we simply impose a tax on these activities, but rather classify them as illegal? Upon deeper reflection, we might come to the conclusion that punishing such activities is largely driven by the societal ethical values rather than financial motives. At this point, however, we start to drift away from the strictly economic theory that views criminal activities and their prevention solely through a financial lens.

1.2 Behavioral Perspectives on Crime

While economics is classified as social science, it diverges to some extent from other disciplines in this field. In its true nature, it is to a large degree purely theoretical as it tries to capture the complex nature of an individual and his role within the society in the form of mathematical formulas and models. Needless to say, to achieve that, it turns a blind eye to a number of facts that form the human psyche and reduces the individual to a *homo oeconomicus*, a self-interested man who thinks rationally in order to maximize his utility. Whereas it sounds great on paper, the harsh reality is that human beings are formed by a great number of external forces that can drastically shift their behavior and these should be accounted for in

empirical analysis.

The ideas of merging the perspective of economic studies with psychology began to appear in the mid 20th century. Until then, human beings were seen as automatons, who respond in the same manner to the same stimuli, as Katona (1951) points out. He argues that there is no meaning in “economics without psychology” neither “psychology without economics” and encourages their intertwinement. By many, he is regarded as the father of the old behavioral economics. Behavioral economics seeks to enrich the classical theories with insight from other social studies and stresses the importance of empirical analysis and data-driven research (Camerer et al., 2004). Its main pillar is the limitation of human rationality, which roots in one’s limited cognitive skills and inability to generate a large amount of possible alternatives and their probabilities and foresee the consequences of future events (Simon, 1955). Individuals often have restricted access to information and little time to make their decision. In addition, they are “manipulated” by their own emotions, which should be, from the traditional economical point of view, irrelevant in their decision-making process.

This is particularly relevant in the study of criminal behavior, as offenders frequently fail to perceive their alternatives objectively. This inconsistency in human behavior can be explained by the concept of prospect theory (Kahneman & Tversky, 1979), which serves as an alternative to the traditional expected utility theory. An individual, rather than assigning value to the final asset, considers his situation compared to a reference point (the current situation he finds himself in) and weights gains and losses. In this process, according to the theory, people tend to overweight small probabilities and underweight large ones, as a result of cognitive bias, suggesting that even a high probability of conviction may be discounted by perpetrators. Besides that, the theory points out the loss aversion – an idea that people are more motivated to avoid losses than to pursue equivalent gains. This implies that individuals who resort to breaking the law may be driven to do so by the desire to escape perceived losses (such as poverty, debt, or social humiliation) rather than the pursuit of financial profit.

As much as individual’s own beliefs are the main driving force in pursuing a “criminal career”, and one has to be held accountable for the consequences of his own actions, it is crystal clear that human consciousness is shaped by the environment in which one is raised. The social system presented to us from an early age defines our ethical principles and therefore it is as well important to look at how individual’s place in the social structure affects his predispositions of becoming an offender.

1.3 The Role of Society

Bartol and Bartol (2017) describe three elementary perspectives on human nature. The first, named *conformity perspective*, views human beings as inherently “good” and aligns closely with positive psychology (a branch of psychology that studies the conditions and processes that have beneficial effect on people and help them thrive). Here, humans are perceived as creatures who want to do the “right” thing and therefore they match their beliefs to the

societal norms. In criminology, this approach is reflected through the *strain theory*. It suggests that humans are generally conforming beings highly influenced by the society in which they live. They desire what other people desire – in many cultures, it is the accumulation of wealth and high status. However, while some can find their ways to success, others find it hard to achieve these goals due to the lack of education, personal contacts, etc. According to this theory, criminal behavior may arise when an individual experiences a discrepancy between the goals cherished by society and his own means to attain them. An individual may feel the urge to become an offender for the sake of keeping up with others.

The second perspective acts as a counterpart to the first one. According to the *nonconformist perspective*, humans lack the feeling of belonging to society and tend to behave in a deviant and undisciplined manner. Since they lack ties to social order, it is necessary for them to be held in check by society; otherwise they would disregard social conventions. Some claim that individuals might incline towards antisocial behavior, such as aggression, due to their biological predispositions presented at birth or acquired later during their formative years. Despite these rather pessimistic views, it is important to note that the nonconformist perspective does not blame people for their deviance. A good example of this perspective is the *social control theory*. Hirschi (1969/2017) hypothesizes, that people are “bad” or at least “imperfect” in nature and what only prevents them from committing crimes are their strong social bonds. If an individual’s attachment to family and friends, commitment to education or career, involvement in conventional activities, or moral beliefs are weakened, there is nothing holding him back from becoming a criminal.

Last but not least, the narrative of the *learning perspective* states that people are born neutral – neither “good” nor “bad” from the start. How they develop is subjective to the environment they live in. To give an example, we present the *social learning theory*. It relies heavily on cognitive processes – thinking, and remembering. Each person creates their version of the world by observing people around him, and social learning theorists advocate that the most important role in the acquisition of human behavior is indeed played by social environment. Therefore, people learn to be criminals because of a message they get from other criminals – as the conventional wisdom says, “bad company promotes bad behavior”. However, there is a chance that individual’s beliefs could be changed; everything depends upon operant conditioning. If he were caught or his mischievous plans were thwarted in other ways, there is a probability that due to developing negative connotations with his activities, he would refrain from such behavior in future, even though this suppression is not expected to be of long-lasting character.

2. Spatial Data Analysis

Socioeconomic data are often inherently linked to geographical locations and can therefore be studied from the perspective of *spatial data analysis*. This approach, naturally, introduces additional information which needs to be taken into account throughout the entire process of exploratory data analysis – the space itself. The spatial component can be conceptualized in various ways; however, in practice, we commonly encounter two representations. Occasionally, data at the level of individual records can be assigned a unique geographic location in the form of coordinates. More frequently, though, statistical surveys collect aggregated data, which refer to predefined territorial units. In such cases, observed values can be assigned to polygons representing these administrative units (municipalities, districts, regions, or perhaps countries). Just as temporal data may exhibit autocorrelation, in spatial data we often encounter *spatial autocorrelation*, which means that the value of a variable at a certain location is correlated with values at nearby locations. This dependence results in the formation of spatial structures and allows us to form clusters of units with similar behavior. Detection of spatial autocorrelation is crucial as it has significant implications for statistical inference and modeling, since it enables us to choose appropriate econometric techniques.

2.1 Spatial Autocorrelation

2.1.1 Proximity

One of the fundamental pillars of spatial data analysis is the determination of proximity. To account for spatial dimension among n cross-sectional units, we can use a matrix of type $n \times n$. The simplest representation is a *binary matrix* $\mathbf{W}[w_{ij}]$, where

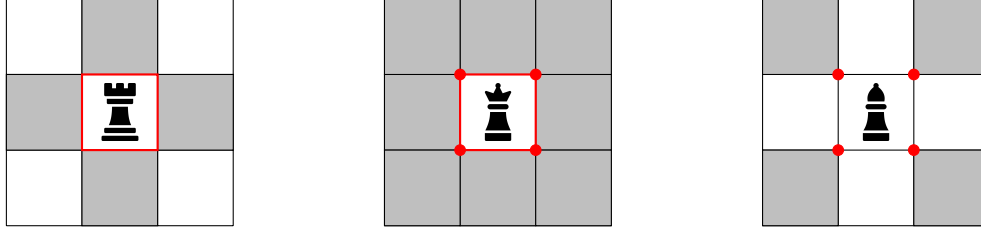
$$w_{ij} = \begin{cases} 1, & \text{if } i \text{ and } j \text{ are contiguous,} \\ 0, & \text{else.} \end{cases} \quad (2.1)$$

For practical purposes, the values of $\mathbf{W}$ are typically scaled, so that the sum of each row equals one. Several methods exist for identifying neighbors, with the most common ones being named after chess pieces, due to the analogy between the graphical representation of contiguity and the movement of these pieces on a chessboard (see Figure 2.1). The *rook criterion* states that two units are neighbors if and only if they share a common border. The extension of the rook criterion is the *queen criterion*, which defines neighboring units as those with either a common border or a common vertex. Finally, the *bishop criterion* considers as neighbors those units that share a single point of contact only, although for its peculiar specification, it is rarely used in empirical analysis.

When determining neighboring, several factors must be taken into account. In the real world, geographical units vary in size and therefore share borders with a differing number

Figure 2.1

Representation of Different Neighboring Criteria (Rook, Queen and Bishop)



Note. Author's own work.

of other units. We could theoretically interpolate the data onto a regular grid, therefore creating a rectangular or hexagonal structure. This would result in most units having an equal number of neighbors, however, the values would lose their original geographical interpretation. Moreover, in some cases, certain units hold special positions that limit their number of neighbors. Examples include coastal or island states in the context of international analysis or districts sharing a border with another country in regional studies.

If, for some reason, we want to address the imbalance in the number of adjacent units among observations, we can use the *k-nearest neighbors* algorithm, where

$$w_{ij} = \begin{cases} 1, & \text{if centroid of } j \text{ is one of the } k \text{ nearest centroids to that of } i, \\ 0, & \text{otherwise.} \end{cases} \quad (2.2)$$

This approach ensures that each observation will be assigned exactly k contiguous units.

A possible technique is to use criteria based on the geographical distance between units. We can use *threshold distance (distance-band weights)*, then

$$w_{ij} = \begin{cases} 1, & \text{if } 0 \leq d_{ij} \leq d_{max}, \\ 0, & \text{if } d_{ij} > d_{max}, \end{cases} \quad (2.3)$$

where d_{ij} is the distance between centroids of units i and j and d_{max} is an established parameter. In this case, two units are considered contiguous if the distance between them is less than or equal to d_{max} . It is crucial to select the appropriate value of the maximum distance parameter, as this method is highly sensitive to the size and structure of observed units. If the threshold is too low, some units may remain isolated. Thus, it is recommended to choose a value that guarantees that each unit has at least one neighbor.

Furthermore, we can construct the *inverse distance matrix* where the weights are defined as a function of the distance between the centroids of observed polygons:

$$w_{ij} = \begin{cases} \frac{1}{d_{ij}^\alpha}, & \text{if } i \neq j, \\ 0, & \text{if } i = j, \end{cases} \quad (2.4)$$

where typically, $\alpha = 1$ or $\alpha = 2$. The links among adjacent units have stronger power and the further the regions are from each other, the weaker their connection is. A slight drawback

of this specification is that the matrix has many non-zero values and with bigger samples, the computational time increases rapidly. Thus, it is generally more convenient to work with sparse matrices. We can, for example apply the inverse distance method to a weight matrix defined by one of the previously mentioned criteria. That is, first determining the contiguity with the use of k -nearest neighbors perhaps, and then differentiating the strength of individual neighbors using the inverse distance technique.

Regardless of the approach used to create the weight matrix $\mathbf{W}$, standardization is generally applied to ensure that the sum of elements in each row is always equal to one, as has already been noted. This can be achieved with a simple transformation:

$$w_{ij}^{stand.} = \frac{w_{ij}}{\sum_{j=1}^n w_{ij}}, \quad i = 1, \dots, n. \quad (2.5)$$

All weights are now between zero and one and the operations with the standardized version of $\mathbf{W}$ can be interpreted as averaging of neighboring values.

2.1.2 Moran's I

Spatial autocorrelation is a phenomenon often present in economic or sociological data and describes a situation in which the realization of a random variable exhibits a systematic spatial pattern. One of its measures is the *Moran's I* statistic initially introduced by Moran (1950), popularized later mainly through the work of Cliff and Ord (1973). It measures the linear association between the value of a given variable, y_i , and the value of this same variable in the neighborhood, y_j (Dubé & Legros, 2014), which can be expressed by the formula

$$I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\sum_{i=1}^n \sum_{j=1}^n w_{ij} \sum_{i=1}^n (y_i - \bar{y})^2}, \quad (2.6)$$

where n denotes the number of observations, y_i and y_j are values of observed variable y in two distinct locations, $\bar{y}$ is the average value of y and w_{ij} is an element of the spatial weights matrix $\mathbf{W}$, which represents the strength of dependence between units i and j . For a row-standardized spatial weights matrix, $\sum_{j=1}^n w_{ij} = 1$, thus formula (2.6) can be rewritten as

$$I = \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} z_i z_j}{\sum_{i=1}^n z_i^2}, \quad (2.7)$$

where z is the original variable in “deviation form”. In most empirical circumstances, Moran's I takes on values from the interval $[-1; 1]$ and its interpretation is similar to that of Pearson correlation coefficient. In the absence of autocorrelation, the statistic is close to zero. Values greater than zero signalize positive spatial autocorrelation, while negative autocorrelation is represented by values less than zero. The more Moran's I diverges from zero, the stronger the autocorrelation.

Given the assumption that y_i follow *iid* normal distribution, the expectation and the variance of Moran's I are given by expressions (2.8) and (2.9):

$$E[I] = -\frac{1}{n-1}, \quad (2.8)$$

$$Var[I] = \frac{n^2 S_1 - n S_2 + 3 S_0^2}{S_0^2 (n^2 - 1)} - (E[I])^2, \quad (2.9)$$

where $S_0 = \sum_{i=1}^n \sum_{j=1}^n w_{ij}$, $S_1 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (w_{ij} + w_{ji})^2$ and $S_2 = \sum_{i=1}^n (\sum_{j=1}^n w_{ij} + \sum_{j=1}^n w_{ji})^2$ (Dubé & Legros, 2014). Assuming normality of y_i , z -score can be calculated:

$$z = \frac{I - E[I]}{\sqrt{Var[I]}}, \quad (2.10)$$

to test for statistical significance of Moran's I .

2.1.3 Local Indicators of Spatial Association (LISA)

The Moran's I gives us an idea about the “average” spatial autocorrelation in our dataset and can be used to test whether our variable is randomly distributed in space or not. However, more often than not, we are interested in closer detail, hence the use of *local indicators of spatial association (LISA)*. The *local Moran statistic* may be computed as

$$I_i = (z_i / m_2) \sum_{j=1}^n w_{ij} z_j, \quad i = 1, \dots, n, \quad (2.11)$$

where, as before, observations z_i and z_j are in deviations from the mean and $m_2 = \sum_{i=1}^n z_i^2 / n$. Furthermore, we can capture the relationship between the local and global version of Moran's I as

$$\sum_{i=1}^n I_i = \gamma I, \quad (2.12)$$

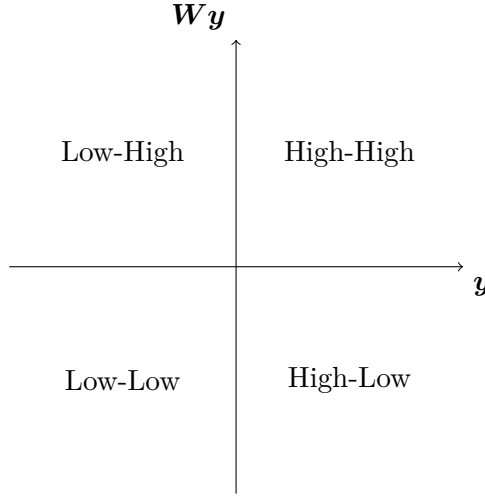
where $\gamma = \sum_{i=1}^n \sum_{j=1}^n w_{ij} m_2$, eventually $\gamma = \sum_{i=1}^n z_i^2$ if $\mathbf{W}$ is row-standardized (Anselin, 1995).

LISA allow us to create heatmaps and identify units with significant levels of spatial autocorrelation. These can be classified into four categories (Dubé & Legros, 2014):

- *High-High*: units fall into this cluster, if their value of the observed variable is high (above average) and they are surrounded by units that also exhibit high values of observed variable,
- *Low-Low*: the exact opposite of *High-High* cluster, consists of units with low (under average) values of observed variable neighboring other units with the same characteristic,
- *High-Low*: also called *diamonds in the rough*, these units present high values, while being surrounded by units with low values,
- *Low-High*: these are units with low values of observed variable surrounded by units with high values of observed variable, also labeled as *black sheep*.

The spatial clusters can also be identified graphically using a *Moran scatterplot*, which describes the relationship between the observed variable of a given unit (on the horizontal axis) and the average value of the same variable at neighboring locations (on the vertical axis). The graph is divided into four quadrants by mean values of observed characteristics. Each quadrant of the plot then represents one of the four clusters (see Figure 2.2 for details).

Figure 2.2
Moran Diagram



Note. Adapted from *Spatial econometrics using microdata*, by Dubé and Legros (2014).

2.2 Spatial Econometric Models

Let us define a classical linear regression model using the equation

$$y = X\beta + \varepsilon, \quad (2.13)$$

where $y \in \mathbb{R}^{n \times 1}$ is a vector of observed dependent variable, $X \in \mathbb{R}^{n \times (k+1)}$ is a matrix of regressors, assumed to be of full rank, $\beta \in \mathbb{R}^{(k+1) \times 1}$ is a vector of parameters, $\varepsilon \in \mathbb{R}^{n \times 1}$ is a vector of unobservable errors (disturbances), n denotes the number of observations, and k represents the number of regressors. We further state the following assumptions about residuals (Greene, 2003; Wooldridge, 2013):

- exogeneity of the independent variables: $E[\varepsilon|X] = \mathbf{0}$,
- homoskedasticity and no serial correlation: $Var[\varepsilon|X] = \sigma^2 I_n$, where $I_n \in \mathbb{R}^{n \times n}$ is an identity matrix,
- normal distribution of disturbances: $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_n)$.

In the most basic specification, it is assumed that the data come from a random sample and individual observations are independent of each other. However, this assumption usually does not hold for data related to geographical units, in which we can often observe spatial patterns. Due to these special characteristics, the classical OLS estimates may become biased and inconsistent, therefore suggesting a different specification of the model, which could address the issue of spatial autocorrelation, might be of bigger interest.

2.2.1 General Nesting Spatial Model (GNS)

Generally speaking, we can identify 3 reasons for spatial interaction effects in data:

1. *Endogenous interaction effects* – dependent variable of one unit depends on the dependent variable of other units,
2. *Exogenous interaction effects* – dependent variable of a unit depends on explanatory variables of neighboring units,
3. *Interaction among disturbances* – error term of one unit depends on error term of other units.

By incorporating all possible interaction effects into a single equation, we get a model

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \mathbf{W} \mathbf{X} \boldsymbol{\theta} + \mathbf{u}, \quad \mathbf{u} = \lambda \mathbf{W} \mathbf{u} + \boldsymbol{\varepsilon}, \quad (2.14)$$

commonly called a *general nesting spatial model (GNS)*, where $\mathbf{W} \mathbf{y}$ represents the endogenous interaction effect, $\mathbf{W} \mathbf{X}$ the exogenous interaction effect, $\mathbf{W} \mathbf{u}$ the interaction among the error terms of different units, $\rho \in \mathbb{R}^{1 \times 1}$ is a spatial autoregressive coefficient, $\lambda \in \mathbb{R}^{1 \times 1}$ a spatial autocorrelation coefficient, $\boldsymbol{\theta} \in \mathbb{R}^{(k+1) \times 1}$ is a vector of unknown parameters and $\mathbf{W} \in \mathbb{R}_+^{n \times n}$ is a non-negative weight matrix defining the spatial configuration of units.

At first glance, this model specification may seem appealing, particularly due to its flexibility; however, it often has a tendency to exhibit weak significance levels of coefficient estimates, which is caused by overparameterization. Consequently, less complex models tend to outperform the GNS model. By imposing restrictions on one or more parameters of the GNS, we can obtain alternative models, which are specified in the following text. The theoretical description of all spatial models presented herein is based on *Spatial econometrics: From cross-sectional data to spatial panels* by Elhorst (2014).

2.2.2 Spatial Error Model (SEM)

When setting $\rho = 0$ and $\boldsymbol{\theta} = \mathbf{0}$, we derive an equation

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} = \lambda \mathbf{W} \mathbf{u} + \boldsymbol{\varepsilon}, \quad (2.15)$$

which defines a *spatial error model (SEM)*. Here, $\mathbf{u} \in \mathbb{R}^{n \times 1}$ is a spatially autocorrelated disturbance vector, while $\boldsymbol{\varepsilon} \in \mathbb{R}^{n \times 1}$ is a vector of independent and identically distributed disturbances. This model is applicable in situations where a potentially important explanatory variable is omitted from the specification and exhibits spatial dependence.

2.2.3 Spatial Lag Model (SLM)

A *spatial lag model (SLM)*, sometimes also called a *spatial autoregressive model (SAR)*, is one of the most commonly used models in empirical spatial modeling. It incorporates spatial process in the response only, which implies setting $\boldsymbol{\theta} = \mathbf{0}$ and $\lambda = 0$, resulting in the equation

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}. \quad (2.16)$$

Interpreting the estimated coefficients is somewhat more complex than in the SEM, due to the existence of both direct and indirect effects. An explanatory variable affects the dependent variable in two distinct ways:

- explanatory variable affects the dependent variable directly – a change in regressor for unit i directly impacts the response in the same unit i ,
- explanatory variable has an indirect effect on dependent variable (also called *spillover effect*) – a change in regressor for unit i has an impact on response in unit j , $j \neq i$ (as a result of spatial lag).

2.2.4 Spatial Lag of Explanatory Variables Model (SLX)

The *spatial lag of explanatory variables model (SLX)*, contrary to the SLM, follows the spatial process in the independent variables, thus $\rho = 0$ and $\lambda = 0$, leading to a model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}. \quad (2.17)$$

This model is considered a local model, as it examines a one-way interaction between a unit and its neighbors. Unlike in the SLM, where the spillover effect is considered global, the SLX assumes that a unit is influenced only by the change in regressors in neighboring locations, which are defined prior to analysis by the weight matrix $\mathbf{W}$. Consequently, the model parameters can be estimated by OLS, similar to a standard regression model. The estimate of $\boldsymbol{\beta}$ captures the direct effects, while the indirect effects are represented by the estimate of $\boldsymbol{\theta}$.

2.2.5 Spatial Durbin Model (SDM)

The *spatial durbin model (SDM)* combines features of both SLM and SLX, meaning the endogenous and exogenous interaction effects are both present, as shown in the equation

$$\mathbf{y} = \rho\mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}. \quad (2.18)$$

The SDM is derived from the GNS by setting $\lambda = 0$.

2.3 Models for Count Data

In empirical research, the dependent variable occasionally takes on non-negative integer values. While the upper bound may or may not exist, the lower bound of the dependent variable is zero. In such cases, the use of OLS is not recommended, as it may lead to negative predictions that are inconsistent with the nature of the regressand. Although the use of count models is often connected with modeling the number of events over specified time intervals, they can be applied in spatial analysis to model the occurrence of events across different geographical units.

2.3.1 Poisson Regression Model

Using the *Poisson regression model* we can model a random variable that follows a Poisson distribution

$$P(Y = y|\lambda) = \frac{e^{-\lambda}\lambda^y}{y!}, \quad y = 0, 1, \dots, \lambda > 0, \quad (2.19)$$

where λ is a parameter of intensity of the process. The distribution models the probability that exactly y events will occur within a given time or across a given space. In the Poisson distribution, $E[Y] = \lambda$ and $Var[Y] = \lambda$, suggesting the equality of mean and variance of the random variable (*equidispersion*).

That is, y_i given $\mathbf{x}_i$ is Poisson-distributed with density

$$f(y_i|\mathbf{x}_i) = \frac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!}, \quad y_i = 0, 1, \dots \quad (2.20)$$

In the log-linear version of the model, the mean parameter is parametrized as

$$\lambda_i = \exp(\mathbf{x}_i'\boldsymbol{\beta}) \quad (2.21)$$

to ensure $\lambda > 0$. Equations (2.20) and (2.21) jointly define the *Poisson log-linear model* (Cameron & Trivedi, 1998).

2.3.2 Negative Binomial Regression Model

In practice, however, the variance of a random variable is often higher in value than the mean. This phenomenon is known as *overdispersion* and can emerge due to several reasons. When the assumption of variance-mean equality is violated, a *negative binomial model* will probably provide a better fit than Poisson model.

Moreover, a situation may arise where an excessive amount of zeros is being produced in the population. If it is believed that the zeros originate from a different data-generating process, implying the existence of two subpopulations, from which one always produces zero count and the other follows a Poisson-like process or negative binomial distribution, the use of a zero-inflated model can be taken into consideration. The *zero-inflated negative binomial model* (*ZINB*) is a combination of two separate models. First, a logit or probit model generates the cases that always produce zeros. Then, a negative binomial model predicts the counts for the rest of the observed units.

3. Methodology and Findings

Building on the theoretical framework introduced in the previous sections, the following part of this thesis focuses on the empirical analysis of criminality and uses the Czech Republic as a study case. It seeks to answer the question of possible spatial autocorrelation of criminal activity, uncover potential problematic geographical areas, and explain differences in crime rates through statistical modeling. Owing to their high severity and expected strong connections to social factors and economic motives, drug-related criminal activities were chosen as the main focus of our analysis. The primary data source for our work was the application *Map of Criminality* administered by the Police of the Czech Republic.

3.1 Open Data of Czech Police

The Czech Police first introduced interactive maps related to criminal activity in November 2016 in response to an ongoing problem regarding the illegal production of methamphetamine and marijuana. The project aimed to inform the general public about the locations of discovered methamphetamine labs and non-industrial cannabis grow houses, primarily in order to raise awareness and protect the health of individuals who might have been affected by such activities. Detailed data collected by the National Drug Headquarters (National Drug Headquarters of the Criminal Police and Investigation Service of the Police of Czech Republic) have since been available for the cadastral areas of municipalities.

Since 2020, the Czech Police in collaboration with Ministry of the Interior of the Czech Republic has been operating an application called *Map of Criminality*, whose main purpose is to disclose information regarding registered criminal offenses committed in the Czech Republic. The available data are structured in such way, that individual records can be accessed. Each observation contains information regarding:

- *type of criminal offense* – classification into one of the statistical categories (violent crimes, burglaries, substance abuse, etc.), in most cases, a subcategory with closer detail is provided as well,
- *time* – day, when the crime was registered into the system,
- *place* – geographical location associated with the offense,
- *relevance* – indicates whether the location is directly connected to the place of offense, or rather to the place of report or consequence of the criminal record,
- *state of criminal offense* – the current status of resolution, providing information on whether the case was solved or not.

The app operates on three distinct levels. The general public only has access to geographically distorted records. This is mainly done to protect personal data of affected individuals and prevent secondary victimization. In addition, municipalities are provided with detailed

statistics that can be utilized to improve the efficiency of local measures in crime prevention. Lastly, the Czech police departments are equipped with datasets and models used to forecast criminal activities in certain areas, to help plan the activities of the police forces.

3.1.1 Data Preparation

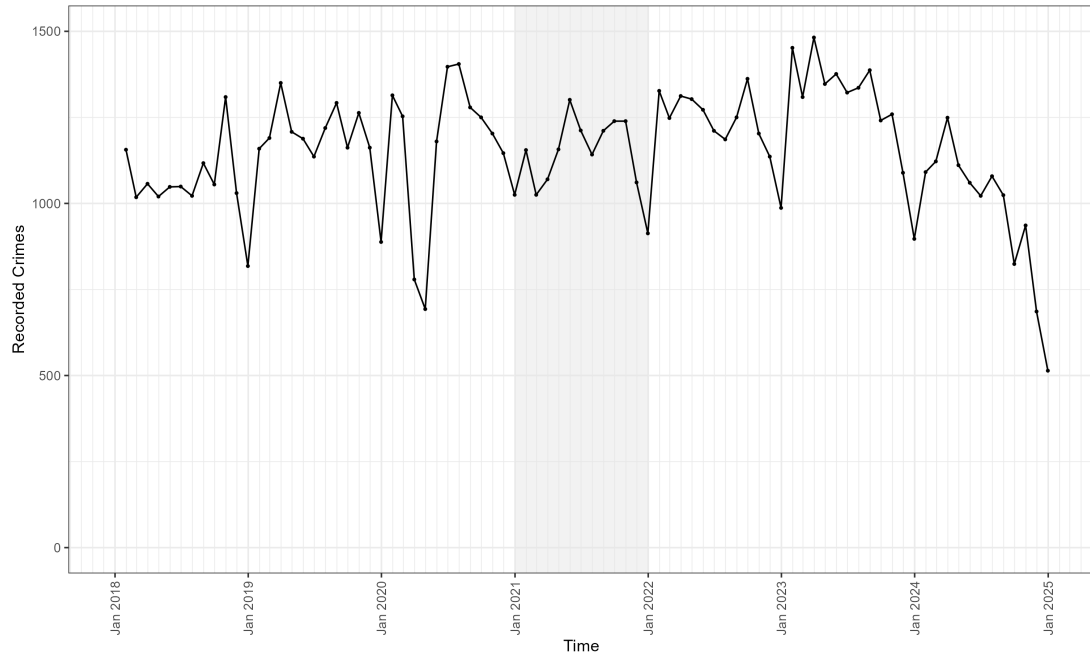
Raw data available to the public (collected from the *Map of Criminality*) were filtered based on several criteria:

- Only drug-related crimes were selected. These fall into three criminological-statistical categories:
 - *Illegal manufacture, possession and distribution of narcotic drugs and poisons* (statistical code: 81),
 - *Spreading of toxicomania* (statistical code: 82),
 - *Other forms of toxicomania, including misdemeanors* (statistical code: 83).
- Dataset was filtered for the time period between January 1, 2018 and December 31, 2024.
- Only observations with place of relevance registered as place of offense were taken into account.
- Records in which it was noted that the offense did not occur were excluded.

To illustrate the dynamics of criminal activity over the monitored period, we present Figure 3.1, which shows the number of criminal offenses aggregated across the entire territory of Czechia. The gray area highlights the year 2021, which will be the focal point of the following analysis. In addition, Table 3.1 shows the annual numbers, providing a detailed breakdown of the share of the three subcategories, as well as the rates of change for the total amount of criminal offenses. We can note that *Other forms of toxicomania, including misdemeanors* have the highest share in the number of drug crimes and, in comparison with the rest, *Spreading of toxicomania* is negligible. The total amount of criminal offenses fluctuates around an average of 13.8 thousand per year and there are no notable seasonal patterns visible at first glance. It is important to note that the observed period was struck by the COVID-19 pandemic. In this context, it is particularly interesting to examine the impact of government-imposed measures on crime levels. During 2020 and 2021, crimes dropped only slightly. While all other categories declined compared to 2019, there was a small increase in misdemeanors in 2020.

3.1.2 Crime Rates

The rest of this thesis is strictly centered around the year 2021, and there is a simple reasoning to this step. To ensure reliable statistical modeling, it is essential to gather a sufficient number of observations. For this reason, we decided to aggregate data concerning criminal

Figure 3.1*Number of Recorded Drug-Related Criminal Offenses in the Czech Republic*

Note. Author's own work. Data sourced from the *Map of Criminality*.

Table 3.1*Absolute Numbers of Recorded Crimes and Their Rates of Change*

Year	81	82	83	Total	Difference	Growth rate (%)
2018	3 164	15	9 520	12 699	—	—
2019	3 517	13	10 687	14 217	1 518	11.95
2020	3 057	9	10 858	13 924	-293	-2.061
2021	3 100	4	10 621	13 725	-199	-1.429
2022	3 110	8	11 679	14 797	1 072	7.811
2023	2 806	13	12 678	15 497	700	4.731
2024	1 686	5	10 027	11 718	-3 779	-24.39

Note. Subcategories presented in the second to fourth column are referred to by their statistical codes (see Chapter 3.1.1 for detailed description).

activities into administrative districts of municipalities with extended powers (AD MEP)¹, in order to acquire 206 observations (205 AD MEP plus the main city, Prague, which holds a special administrative status). However, obtaining additional statistical data at this level of granularity can be challenging, as such fine-grained information is usually not available on an annual basis. Nevertheless, the recent population census, conducted in 2021, provides a rich source of demographic data at the level of municipalities and city districts. Therefore, the year 2021 was selected as the reference period, since it is the closest period to this day, for which all the necessary data are available.

Figures 3.2 and 3.3 show individual records – raw data for year 2021 obtained from the police database. Due to limited number of records, incidents classified as *spreading of toxicomania* are not visualized. However, for the sake of completeness, we add that these incidents were recorded in AD MEP Brno, Jihlava, Roudnice nad Labem and Žatec. These visualizations do not provide much insight, except for the rather intuitive observation that more incidents tend to happen in places where there are more people. That is why, we can see a higher concentration of crime in the vicinity of larger cities. Given that administrative units differ in terms of population size, it is necessary to compute crime rates to ensure correct interpretation.

From this point onward, the analysis is limited exclusively to criminal offenses classified as *Illegal manufacture, possession and distribution of narcotic drugs and poisons*. Offenses falling under the remaining two categories are entirely excluded from consideration. While category 82 includes only a very limited number of cases, category 83 is quite ambiguous, as it consists of both drug-related offenses and cases unrelated to narcotics. Specifically, it consists of cases of *Endangerment under the influence of an addictive substance* and *Drunkenness*, however, no further detail is provided that would allow for a clear distinction between these two subcategories. Given the objective of our analysis is to examine drug-related criminal behavior particularly, we decided to exclude criminal offenses from this category due to their rather vague description to ensure interpretation clarity. Hence, crime rates are relative only to the statistical category numbered 81.

Figure 3.4 displays the number of selected criminal offenses committed in 2021 per thousand inhabitants. Population sizes are based on the *2021 Population and Housing Census*. AD MEP Nový Bor, Jeseník, Náměšť nad Oslavou and Bruntál belong among the units with the highest crime rates, each with over 0.8 drug crimes per thousand inhabitants.

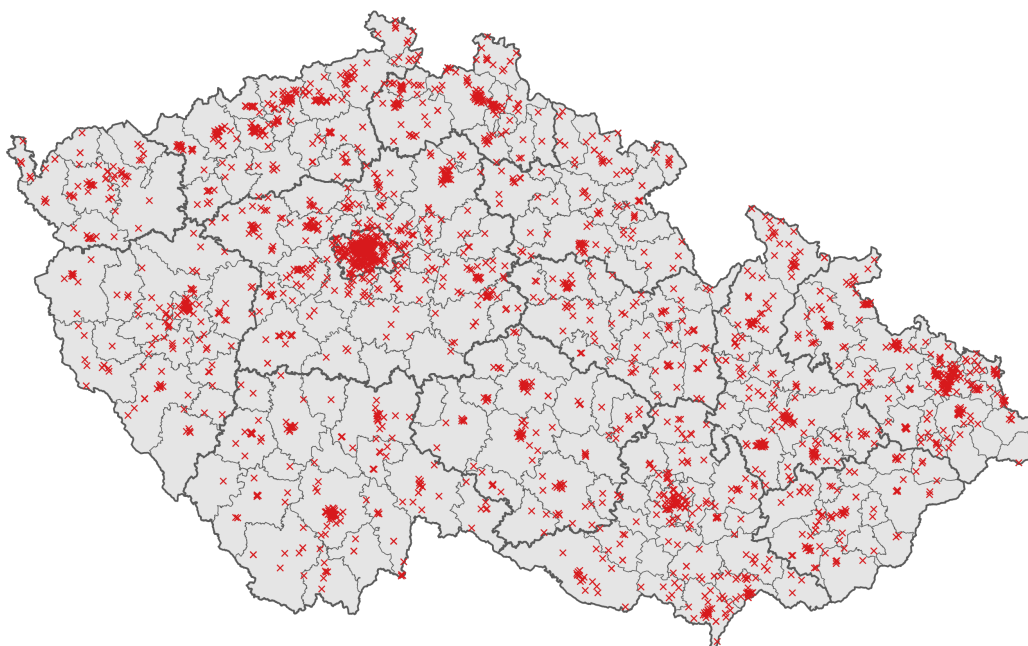
3.2 Analysis of Spatial Dependence

The visualization of crime rates in administrative units reveals a certain pattern in the spatial distribution of this statistic. In the section that follows, we therefore seek to test the signif-

¹Note. The original Czech term is *správní obvody obcí s rozšířenou působností (SO ORP)*.

Figure 3.2

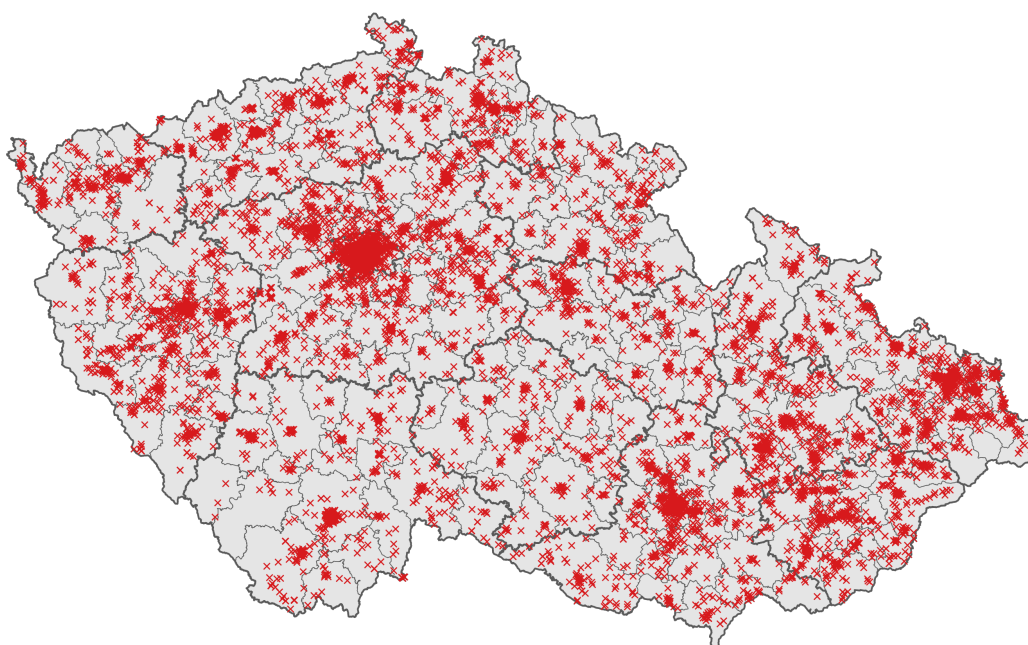
Records of Illegal Manufacture, Possession and Distribution of Narcotic Drugs and Poisons



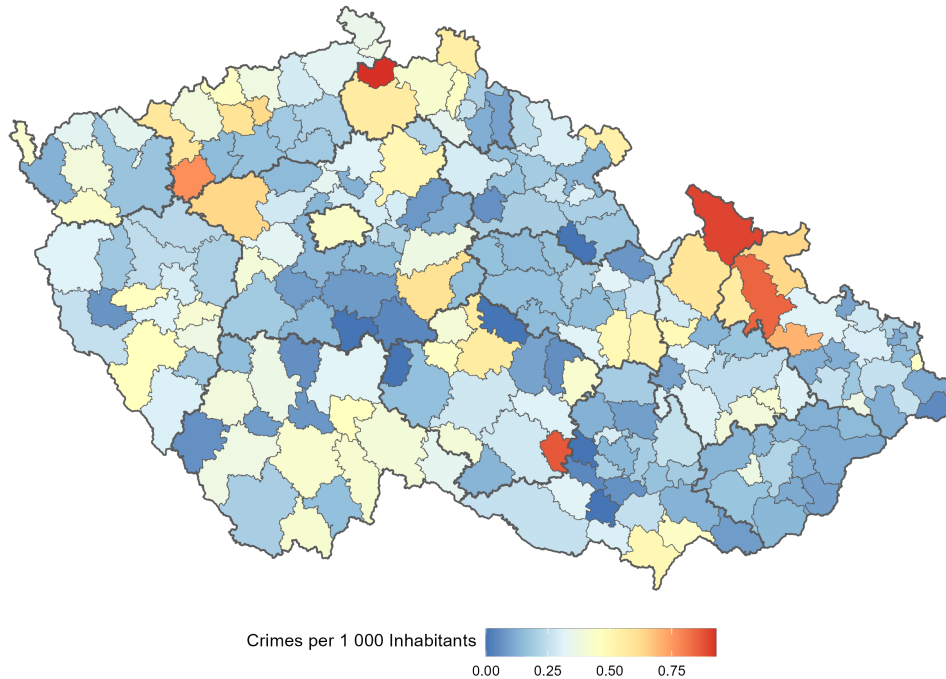
Note. Author's own work. Data sourced from the *Map of Criminality*.

Figure 3.3

Records of Other Forms of Toxicomania Including Misdemeanors



Note. Author's own work. Data sourced from the *Map of Criminality*.

Figure 3.4*Crime Rate in the Czech Republic*

Note. Author's own work. Data sourced from the *Map of Criminality* and the *2021 Census*.

icance of spatial autocorrelation in this variable. However, we first need to mathematically determine the neighboring structure of observed areas and create a weight matrix.

3.2.1 Weight Matrices

Given the social character of our data, it is beyond one's analytical skills to correctly guess all the underlying processes that influence the formation of spatial relationships between units. Surely, geographical location and distance play significant roles; however, the strength of adjacency may also be affected by the level of infrastructure, length of common border, commuting patterns, etc.

Nevertheless, the right construction of the spatial weights matrix plays a crucial role, as different definitions of neighboring can change the results of statistical tests and models. Since we cannot determine with certainty which configuration of proximity is most appropriate, in the next section, we therefore present several neighboring matrices computed based on different criteria.

Figures 3.5 and 3.6 show neighboring relations based on common border. The queen and rook criterion do not differ much in the case of Czechia, since the only units sharing just one point of contact are Brno and Blansko. Both weight matrices are composed in such way that

gives individual neighbors the same importance, in other words, by standardizing the binary proximity matrix so that each row sums up to one. On average, in the queen configuration, each AD MEP has 5.39 neighbors.

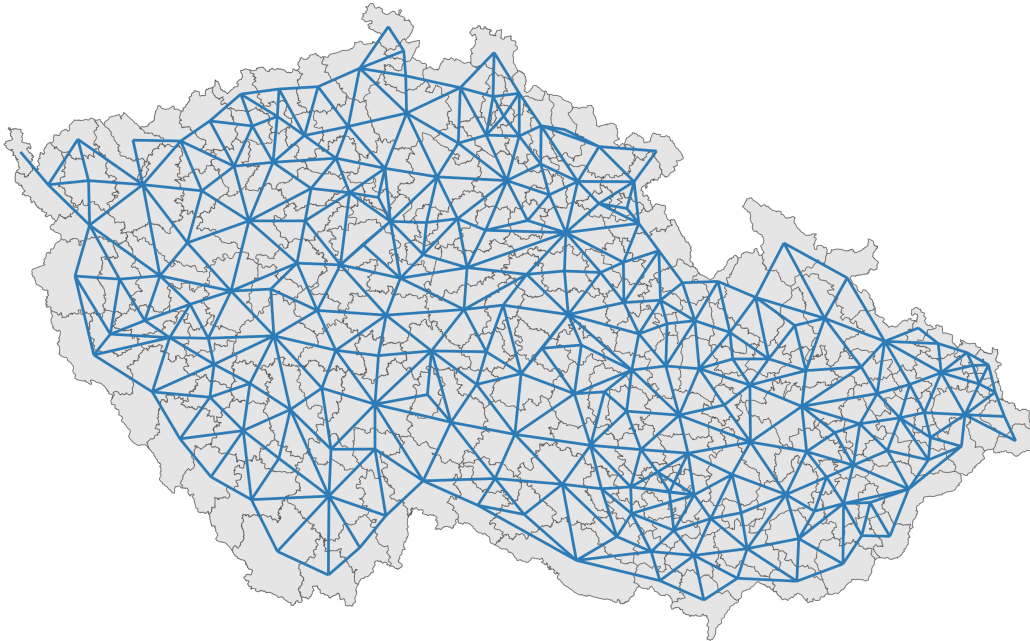
Next comes the definition of adjacency based on the k -nearest neighbors algorithm, where we chose k to be equal to five and the geometrical centers of polygons were selected to be the centroids of units. The final weight matrix was again standardized in the same way as in the previous two cases. We present the visualization in Figure 3.7.

Lastly, we compute the proximity matrix with the use of threshold distance (see Figure 3.8). Two units satisfied the condition of neighboring, if the distance between their centroids did not exceed 50 kilometers. According to this configuration, each AD MEP has 17.15 neighbors on average. Regarding the high average number of adjacencies, the weight matrix was constructed as an inverse distance matrix, to give a higher level of importance to closer units, and only then was the matrix row-standardized.

The various configurations of geographical units are to be used and compared in order to find out whether they change the results of statistical tests.

Figure 3.5

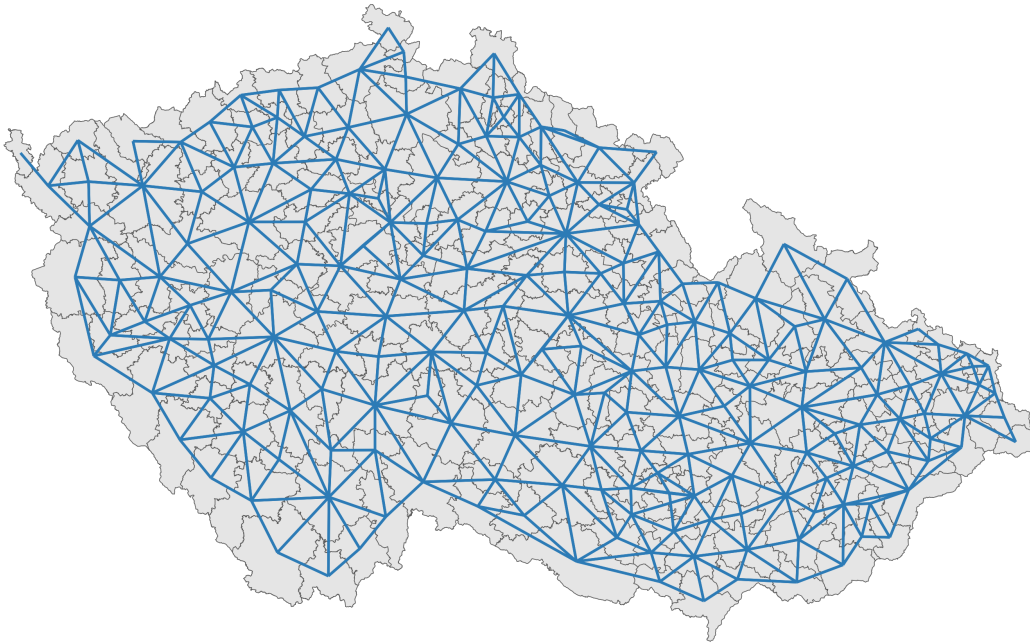
Definition of Adjacency Based on the Queen Criterion



Note. Author's own work.

Figure 3.6

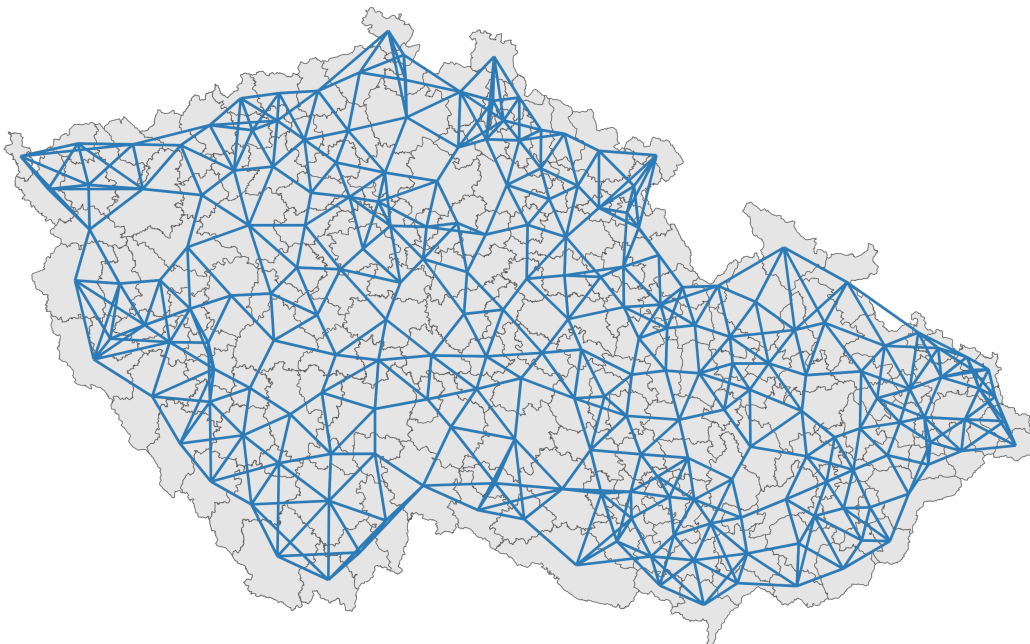
Definition of Adjacency Based on the Rook Criterion



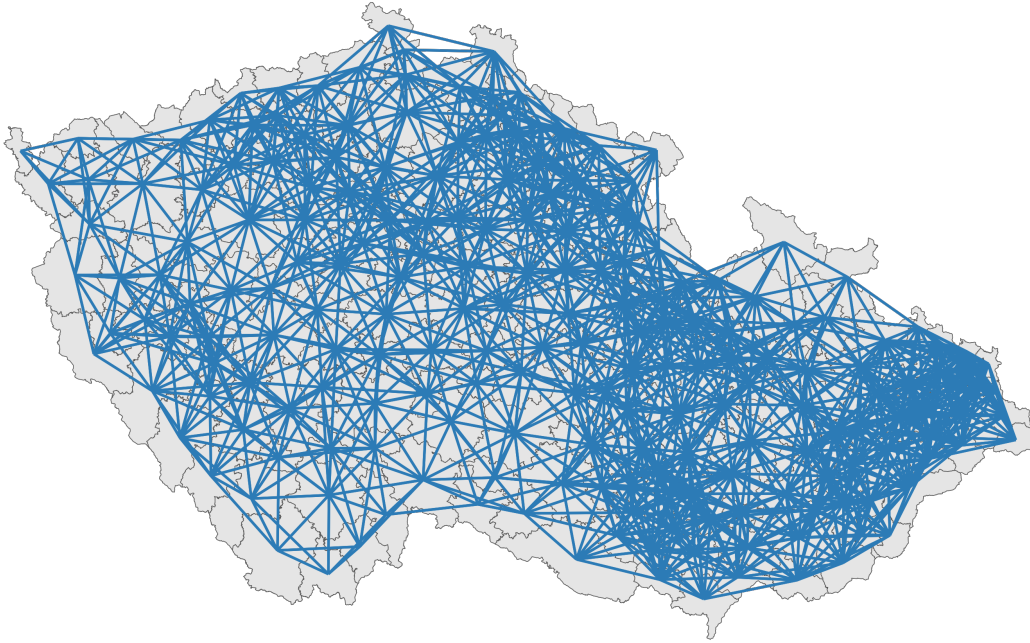
Note. Author's own work.

Figure 3.7

Definition of Adjacency Based on k -Nearest Neighbors



Note. Author's own work.

Figure 3.8*Definition of Adjacency Based on Threshold Distance**Note.* Author's own work.

3.2.2 Moran's I

To test for spatial autocorrelation in data, we first calculate the value of global Moran's I statistic and then test the following hypothesis

$$H_0 : I = 0$$

against its one-sided alternative

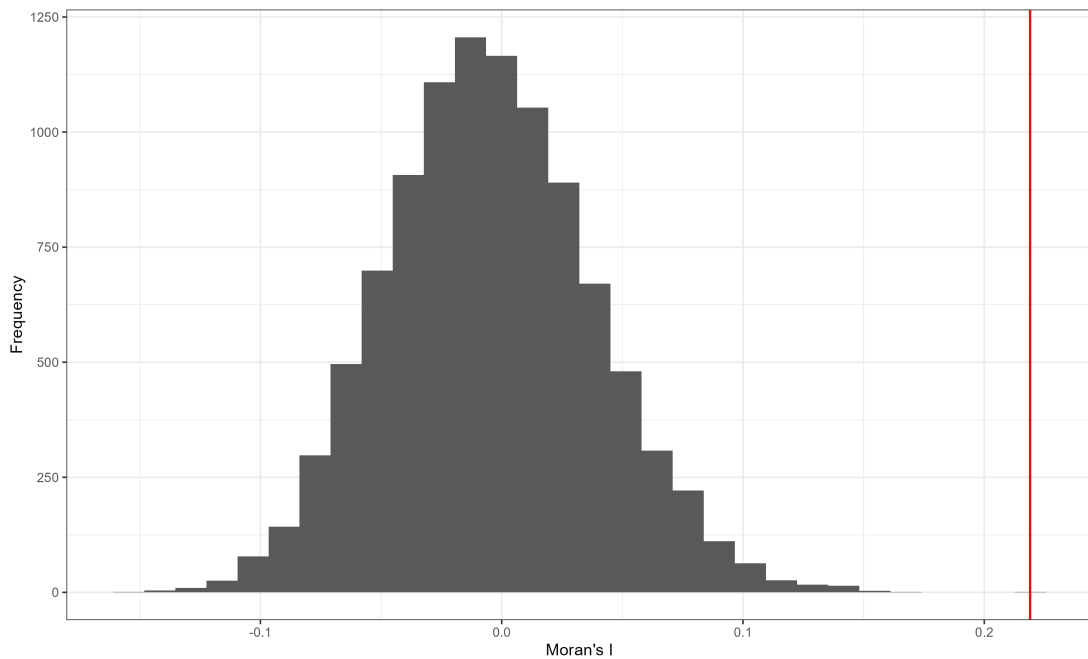
$$H_1 : I > 0.$$

Under the assumption of normality, the Moran's I statistic is asymptotically normally distributed (Cliff & Ord, 1981) and the results of a statistical test may be obtained by comparing the test statistic with standard normal distribution. In practice, we test the significance of Moran's I with a permutation test. This approach makes use of Monte Carlo simulation. The observed values are "shuffled" and randomly assigned to geographical units many times, and the Moran's I statistic is calculated for each permutation. This gives us the empirical distribution of I under the null hypothesis of no spatial autocorrelation. We can then compare the actual value of Moran's I with this distribution to obtain a pseudo p -value. The results of the permutation test are shown in Table 3.2. Figure 3.9 displays the empirical distribution of I under H_0 obtained from 999 permutations, where the red line marks the observed value of Moran's I . We reject the null hypothesis of no spatial autocorrelation for all listed weight matrices. Therefore, we state that crime rate is positively spatially autocorrelated, which implies that areas with high levels of criminality are likely to be surrounded by other areas with high criminality and vice versa.

Table 3.2*Moran's I Permutation Test*

Weight matrix	Moran's I	Pseudo p -value
Rook criterion	0.2184	0.0001
Queen criterion	0.2190	0.0001
5 nearest neighbors	0.1930	0.0001
Distance-band (up to 50 km)	0.1553	0.0001

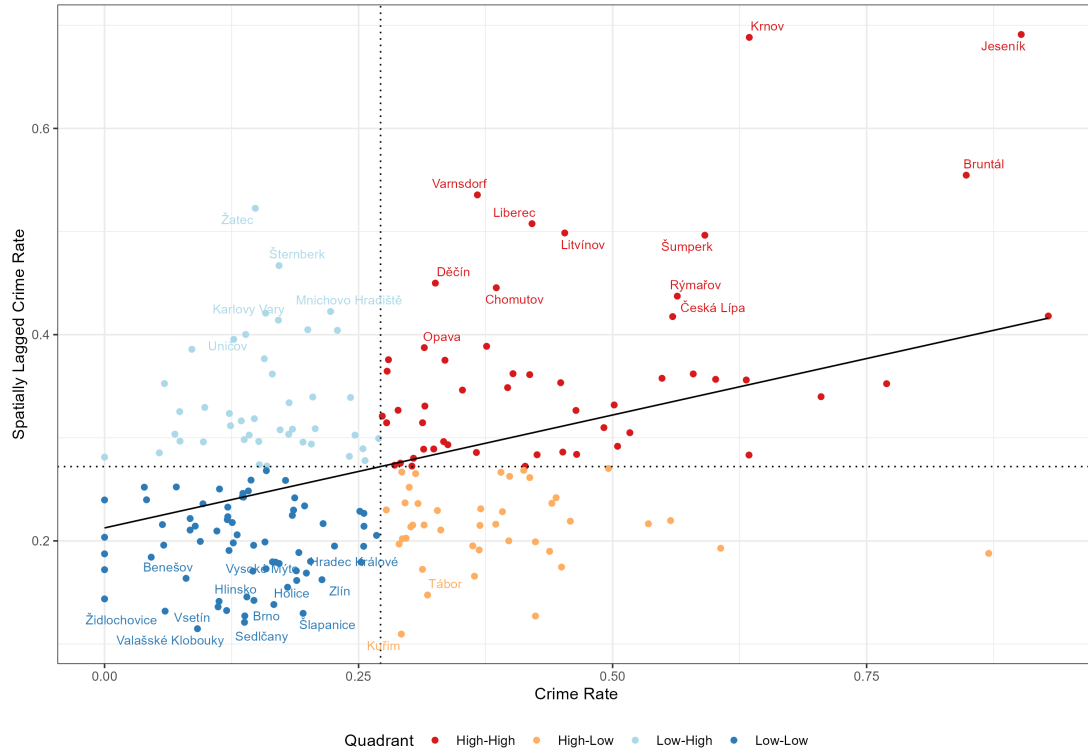
Note. (i) Computed using Monte Carlo simulation with 9 999 random permutations. (ii) Distance-band matrix weights calculated with the use of inverse distance, the rest uses classical row-standardization of binary matrices.

Figure 3.9*Results of Moran's I Permutation Test*

Note. Author's own work. Data sourced from the *Map of Criminality* and *2021 Census*.

The distribution of the geographical units is illustrated in the Moran scatterplot (see Figure 3.10). It shows the classification of AD MEP into four clusters based on their crime rates in relation to their spatial lag. More specifically, it compares the crime rate in a certain location with the average crime rate in neighboring areas. The plot is constructed using the queen criterion to define adjacency units. Since most geographical units fall into the first or third quadrant, the spatial dependence is positive. The slope of the regression line is an estimation of the global Moran's I .

Figure 3.10
Moran Scatterplot



Note. Author's own work. Data sourced from the *Map of Criminality* and the *2021 Census*.

3.2.3 LISA

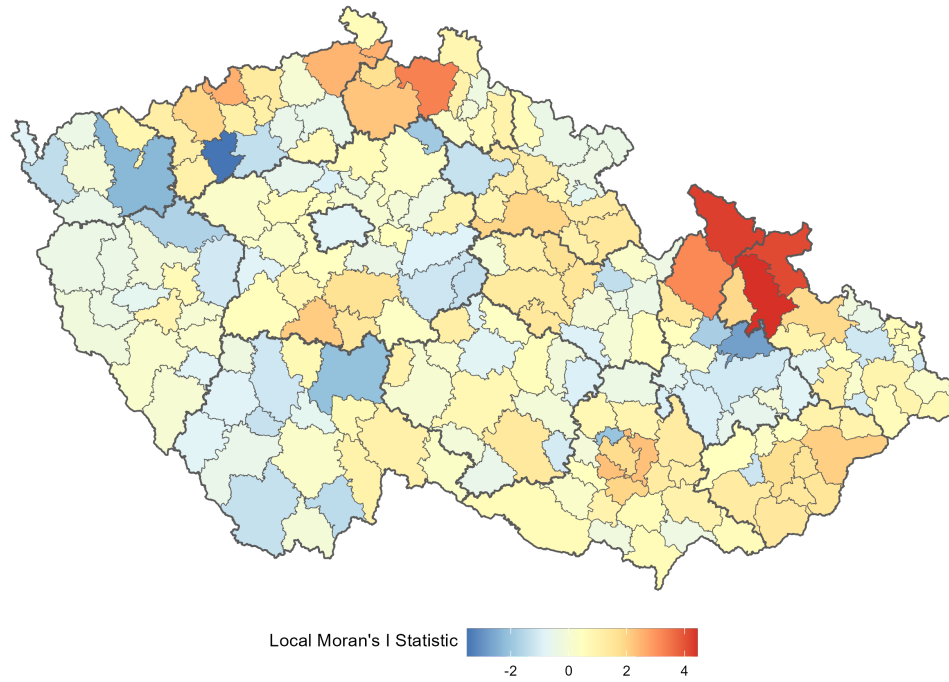
Figure 3.11 captures the standardized version of local Moran's I statistic for each observation. Values above ≈ 1.96 or below ≈ -1.96 indicate statistical significance at the 5% level. While positive values of Z -score suggest that the area belongs to a High-High or Low-Low cluster (grouping of similar characteristics), negative values imply that crime rate for a given location differs considerably from those of its neighbors (indicating classification into High-Low or Low-High cluster). The final clusters are displayed in Figure 3.12 – this visualization takes into consideration levels of crime rate in geographical units as well as the differentiation between statistically significant and insignificant areas. Most AD MEP belong to the cluster of insignificant units. The analysis uncovered three main hotspots, all of which are located in the northern or northeastern regions of Czechia. These clusters are located within the Ústí nad Labem, Liberec, Olomouc and Moravian-Silesian regions.

3.3 Analysis of Socio-Economic Factors

In the next section, we take a look at a number of selected socioeconomic characteristics and focus on examining their strength of association with crime rate. The set of chosen

Figure 3.11

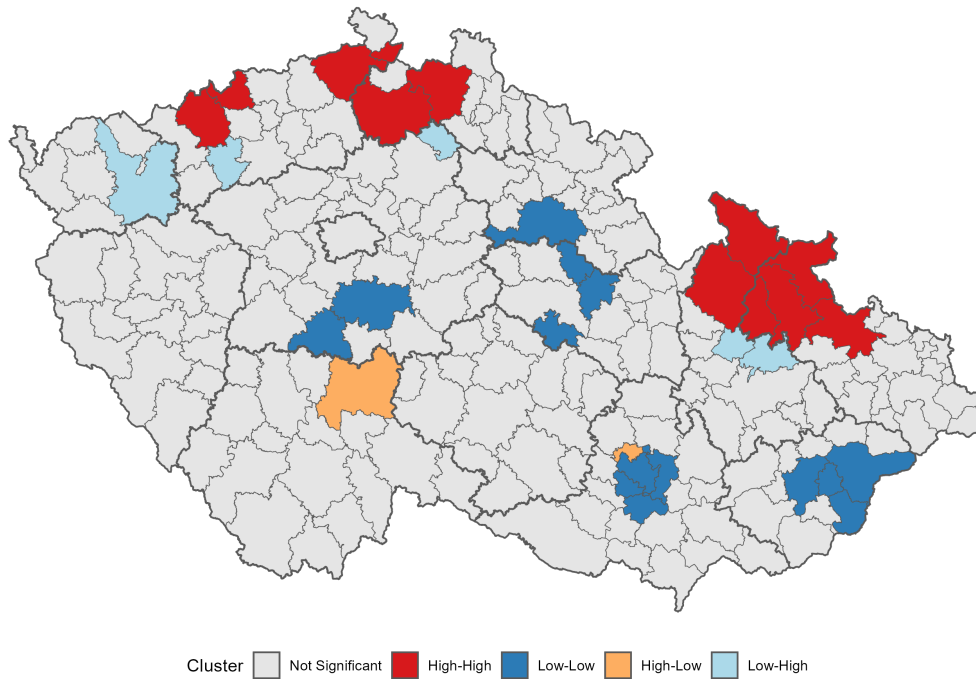
Local Moran's I Statistic (Standardized)



Note. Author's own work. Data sourced from the *Map of Criminality* and the *2021 Census*.

Figure 3.12

LISA Clusters



Note. Author's own work. Data sourced from the *Map of Criminality* and the *2021 Census*.

variables includes both demographic characteristics of inhabitants and indicators reflecting their economic situation.

3.3.1 Age and Sex Structure

The main two characteristics concerning demography are sex and age. Instead of including several variables that would capture the whole population structure, we chose to work with a variable, which represents the percentage of men aged 15 to 34 years in the total population. This variable is inspired by the so-called *age-crime curve* (Hirschi & Gottfredson, 1983), which illustrates the frequency of committing criminal offenses across all ages for both sexes separately. This curve has an asymmetrical bell shape, indicating that criminal activity increases sharply during adolescence and, after reaching its peak in the late teenage years or early twenties, tends to decline. In addition, frequencies for men are significantly higher than for women. Therefore, the chosen variable should reflect the proportion of population most prone to engaging in criminal behavior – the group that contributes most to the increase in crime.

3.3.2 Foreigners

A potentially interesting relationship to study is that between crime rates and foreigners. In this context, different studies present different findings, and while some support the idea that immigrants contribute significantly to crime statistics, others find no such evidence. Our idea is to track the percentage of foreigners in the population in order to find out whether an increase in people of foreign nationality has an impact on the amount of drug-related activities. One possible link between these variables may lie in the activities of organized criminal networks, which often operate on international levels. Foreigners may be involved in the trafficking of illegal substances or precursors across borders and drug distribution. Another perspective may consider the possibility that some immigrants may face difficulties integrating into society and securing stable employment. Faced with a lack of financial resources and living in marginalized areas, they can resort to breaking the law to ensure their well-being.

3.3.3 Low Education

Particular attention is as well given to examining the proportion of people with low education within the areas of interest. This variable captures the percentage of people without any formal education as well as those whose highest attained education is primary education. A negative relationship between the level of education and criminality can be intuitively explained by the increased probability of unemployment among undereducated individuals. Even when employed, they are more likely to have low-income jobs. As a result, they may experience material deprivation, which can open up a path to financially motivated crimes

(such as theft, fraud, or drug distribution). Moreover, these individuals may perceive the potential loss stemming from unlawful behavior as relatively low, since they have little to lose anyway. The debate on the slight limitation of cognitive abilities in this group is also pertinent – people lacking schooling may struggle to recognize negative impacts criminality has on their lives and society, as well as to properly assess the risks and consequences of their actions. These behavioral patterns are, in many cases, transmitted in families to younger generations, contributing as well to a rise in juvenile delinquency.

3.3.4 Divorce Rate

The choice to include a variable that reflects divorce rate may not be immediately apparent. However, divorce is a potent risk factor for the onset of drug abuse. Adults may turn to drugs as a coping mechanism to deal with the stress and anxiety associated with family breakdown. Emotional distress and family instability especially affect children, who may lack parental supervision and support, rising their risk of dropping out of school, engaging in delinquency and falling into addiction. This early exposure to criminal world can evolve into participation in drug production and distribution. Some children from single-parent households may get into drug business to try to compensate for the missing parent's income, while others may get involved simply to fuel their own addiction. Conversely, the relationship between the variables may also operate in the opposite direction – divorce may occur as a consequence of substance abuse of one of the family members.

3.3.5 Unemployment Rate

High unemployment rate is closely associated with financial instability and social exclusion. The lack of legal employment opportunities encourages individuals to seek alternative sources of income. Prolonged unemployment causes psychological distress, feelings of hopelessness, and loss of social status, thereby increasing the propensity to engage in illegal activities. Beyond that, as noted in the previous text, individuals may become involved in drug trade as a consequence of their own addiction. Such abuse of addictive substances is often initiated and exacerbated by adverse life circumstances, including, for instance, the loss of employment.

3.3.6 Financial Security

Even though some of the indicators mentioned so far have the ability to partially reflect the economic well-being of the population, our aim was to incorporate a variable that would directly capture the wealth of the population. The initial idea was to include information on wage differences in Czechia; however, data availability is a major obstacle here. A similar issue arises with regard to data concerning material deprivation, as both are available only at regional levels. For this reason, we chose two alternative variables that could be associated with the level of poverty and financial desperation. The first variable measures the number

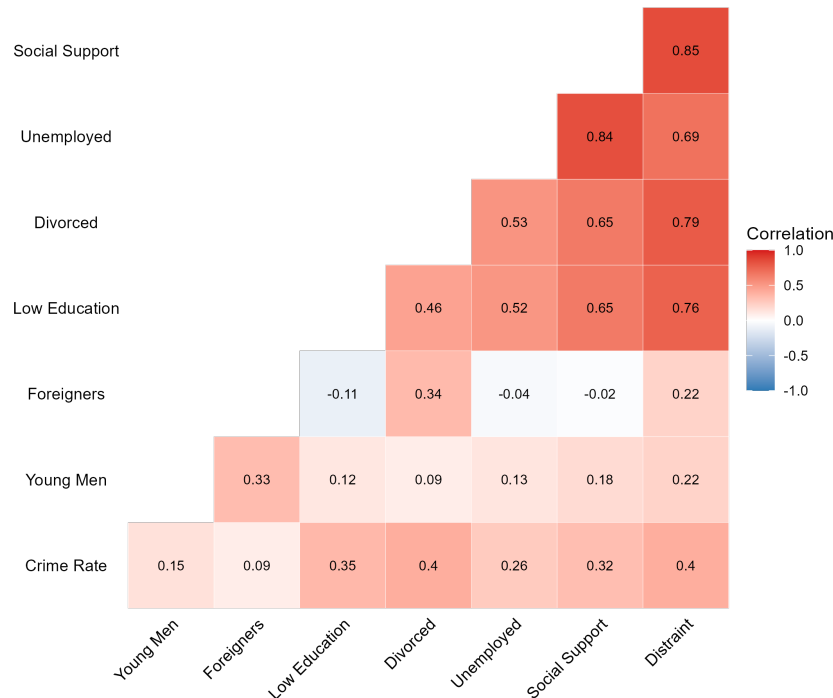
of allowances for living per 1 000 inhabitants over the age of 15 (values represent monthly averages for a given year), while the second measures the percentage of individuals subject to debt enforcement proceedings (distrain) in population.

Data concerning financial security were retrieved from *DataPAQ*. The unemployment rates were calculated as an average share of attainable jobseekers among the population in productive age (15–64) through the twelve months of 2021 based on statistics published by the *Ministry of Labour and Social Affairs*. The rest of the demographic data were retrieved from the *2021 Population and Housing Census*.

Correlogram in figure 3.13 illustrates the strength of linear dependencies among the pairs of variables under consideration. With regard to the set of variables related to financial security and their high degree of mutual correlation, we opted for the variable measuring the number of living allowances per 1 000 residents above the age of 15 to be included in regression models rather than the share of individuals subject to debt enforcement proceedings. Figure 3.14 provides choropleth maps of all six selected regressors. Due to the skewed distributions of the unemployment rate, the share of foreigners and the relative number of benefits in material need, logarithmic transformation will be applied to these variables during the statistical modeling process.

Figure 3.13

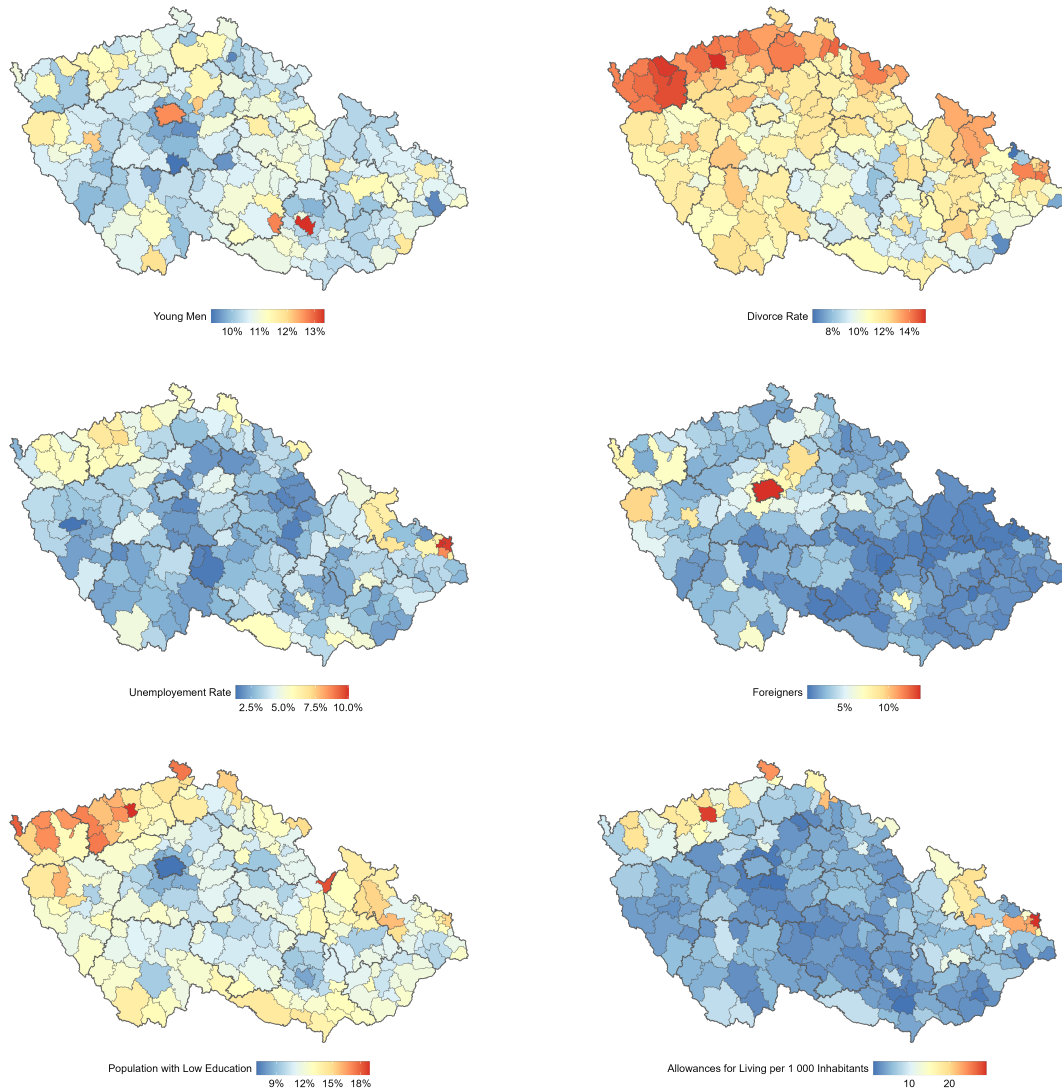
Correlogram



Note. Author's own work. Data sourced from the *Map of Criminality*, *2021 Population and Housing Census*, *Ministry of Labour and Social Affairs*, and *DataPAQ*.

Figure 3.14

Choropleth Maps of Independent Variables



Note. Author's own work. Data sourced from the *2021 Population and Housing Census*, *Ministry of Labour and Social Affairs*, and *DataPAQ*.

3.4 Econometric Models

3.4.1 Classical Linear Regression

We begin the analysis with a classical linear regression model defined by the following equation:

$$\begin{aligned} crime_rate = & \beta_0 + \beta_1 young_men + \beta_2 divorced + \beta_3 \log(unemployed) \\ & + \beta_4 \log(foreigners) + \beta_5 low_education + \beta_6 \log(social_support) + \varepsilon. \end{aligned}$$

The coefficient estimates are reported in Table 3.3. Based on the Breusch-Pagan test, the null hypothesis of homoskedasticity is rejected (p -value = 0.0125), and robust standard errors are provided in the regression results. The findings indicate a significant positive effect of the divorce rate and the share of people with low education in the population on the crime rate. In contrast, the level of social support is associated with a statistically significant negative effect. The impact of the remaining variables is found to be insignificant at the 5% level. The model accounts for 22.69% of the variability in the dependent variable. All variance inflation factors are below five, signaling no signs of multicollinearity. The Monte Carlo permutation test applied to the residuals reveals the presence of spatial dependence (p -value = 0.0030) and points to the possibility of expanding the model with a spatial component.

Table 3.3

Results of Classical Linear Regression Explaining Number of Crimes per 100 000 Inhabitants

	Linear Regression
Young Men	4.920 (2.933)
Divorced	5.433*** (1.255)
Log Unemployed	3.924 (4.127)
Log Foreigners	−5.142 (2.939)
Low Education	1.951** (0.6862)
Log Social Support	−7.263* (3.616)
Intercept	−98.42** (32.81)
<i>N</i>	206

Note. (i) Robust standard errors are given in parentheses. (ii) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

3.4.2 Spatial Econometric Models

To assess whether, and which forms of spatial interaction should be incorporated into the classical linear regression model, Lagrange Multiplier (LM) tests are employed. Both *LMerr* and *LMlag* evaluate the null hypothesis of the absence of spatial autocorrelation in the error term ($\lambda = 0$) and in the dependent variable ($\rho = 0$), respectively. The null hypotheses are rejected for both *LMerr* (p -value = 0.0028) and *LMlag* (p -value = 0.0030) tests. Following the algorithm proposed by Anselin et al. (1996), the diagnostic testing is extended by applying robust versions of the LM tests, which evaluate the presence of spatial error process while controlling the possible presence of spatial lag, and vice versa. However, a rather rare instance occurs, wherein neither the robust *LMerr* (p -value = 0.5560) nor the robust *LMlag* (p -value = 0.6811) yields statistically significant results, suggesting that more serious specification problems are likely present. Despite the results, both spatial models are estimated, and the analysis is further broadened to include the SLX and SDM specifications. Table 3.4 presents the coefficient estimates; the discussion and comparison of the models follows in Chapter 3.4.4.

3.4.3 Poisson Regression Model and Negative Binomial Regression Model

Addressing the nature of the dependent variable, the focus is further directed towards the use of count models to capture the associations between selected socioeconomic factors and criminality. Since we have access to records of individual criminal offenses (and thus aggregated counts for all administrative units), the dependent variable can be treated as a count variable. However, given that the offense counts are not directly comparable across the observed units, this disparity must be considered in the model specification. This is done by introducing exposure into the model, which is in our case represented by the population size, as it is reasonable to observe higher levels of criminality in more populated areas.

At first, we estimate the Poisson model:

$$\begin{aligned} \log(\text{crimes}) = & \beta_0 + \beta_1 \text{young_men} + \beta_2 \text{divorced} + \beta_3 \log(\text{unemployed}) + \beta_4 \log(\text{foreigners}) \\ & + \beta_5 \text{low_education} + \beta_6 \log(\text{social_support}) + \log(\text{population}), \end{aligned}$$

where $\log(\text{population})$ serves as an offset. The results of this regression are shown in Table 3.5. To assess the suitability of the Poisson model, we conduct a simulation-based test for overdispersion. Based on the test result, we reject the null hypothesis (p -value = 0.0000), which signifies the violation of the equidispersion assumption inherent to the Poisson model. Consequently, we estimate the negative binomial model, which introduces an additional parameter to account for the unobservable heterogeneity. The results of this regression are reported in Table 3.5. After repeating the dispersion test for the NB model, we do not find sufficient evidence to reject the null hypothesis (p -value = 0.0782). Furthermore, we apply a simulation approach to check for zero-inflation; however the outcome does not indicate a

Table 3.4*Results of Spatial Models Explaining Number of Crimes per 100 000 Inhabitants*

	SEM	SLM	SLX	SDM
Young Men	5.211* (2.084)	5.275** (2.041)	5.141* (2.275)	5.540** (2.131)
Divorced	5.140*** (1.208)	4.900*** (1.173)	5.021*** (1.378)	4.825*** (1.292)
Log Unemployed	2.692 (4.631)	3.030 (4.260)	2.649 (5.402)	1.749 (5.063)
Log Foreigners	-4.015 (2.779)	-4.654 (2.419)	-1.700 (3.878)	-1.534 (3.635)
Low Education	2.224** (0.7741)	1.663* (0.7162)	2.073* (1.026)	2.228* (0.9614)
Log Social Support	-6.780 (3.638)	-7.066* (3.411)	-7.631 (4.179)	-7.518 (3.920)
Intercept	-101.9*** (26.18)	-99.89*** (25.29)	-109.2 (64.98)	-97.39 (61.78)
Lag – Young Men			-1.198 (5.091)	-1.969 (4.791)
Lag – Divorced			3.798 (3.055)	3.239 (2.943)
Lag – Log Unemployed			9.456 (9.295)	10.59 (8.745)
Lag – Log Foreigners			-9.763 (5.749)	-9.047 (5.437)
Lag – Low Education			-0.6721 (1.552)	-1.572 (1.461)
Lag – Log Social Support			-8.532 (7.901)	-8.519 (7.491)
Lambda	0.2917**			
Rho		0.2770**		0.3094**
N	206	206	206	206

Note. (i) Standard errors are given in parentheses. (ii) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

significant issue with an excess of zeros (p -value = 0.6854), thus the ZINB model is deemed unnecessary.

Table 3.5

Results of Poisson Regression and Negative Binomial Model Explaining Crime Rate

	Poisson	NB
Young Men	0.1254*** (0.03278)	0.1531* (0.07162)
Divorced	0.1268*** (0.02281)	0.1946*** (0.04313)
Log Unemployed	-0.03209 (0.08894)	0.06979 (0.1617)
Log Foreigners	0.08509* (0.04306)	-0.1244 (0.08399)
Low Education	0.04142*** (0.01182)	0.05644* (0.02457)
Log Social Support	-0.1546* (0.07059)	-0.2365 (0.1255)
Intercept	-11.28*** (0.4473)	-12.34*** (0.8999)
N	206	206

Note. (i) Standard errors are given in parentheses. (ii) * $p < 0.05$,
** $p < 0.01$, *** $p < 0.001$

3.4.4 Model Comparison

To select among several competing models, one can utilize the natural logarithm of the likelihood function $\mathcal{L}$, as higher log-likelihood values indicate an overall better fit. Additional essential tools for model comparison are information criteria, such as the *Akaike information criterion* and the *Bayesian information criterion*, both of which are derived from the value of the likelihood function. The former is defined as

$$AIC = -2\ln(\mathcal{L}) + 2k, \quad (3.1)$$

where k denotes the number of fitted parameters (including the intercept). The latter is defined as

$$BIC = -2\ln(\mathcal{L}) + \ln(n)k, \quad (3.2)$$

where n represents the sample size. Both criteria reward goodness-of-fit but penalize excessive complexity. Notably, the BIC tends to favor more parsimonious models. Both criteria are minimizing.

The values of Akaike and Bayesian information criteria for all estimated models are presented in Table 3.6 along with the log-likelihood values. Among all models, the negative binomial model performs best across all three criteria and is therefore selected as the preferred choice. In comparison, the Poisson model yields poorer results, as it fails to account for the overdispersion present in the data. Although the classical linear regression model was used as the starting point of the analysis, it has been outperformed by all the other models in terms of goodness-of-fit as indicated by the log-likelihood values. The SEM, SLM, and SDM yield better results based on the Akaike information criterion, and while this suggests that accounting for spatial dependence provides a slight enhancement, the improvements are only marginal. The SLX achieves the highest AIC, likely due to its large number of parameters introduced by including spatial lags of all explanatory variables. This increased complexity becomes even more evident when considering the Bayesian information criterion, which penalizes additional parameters more strictly. A similar pattern can be observed in the SDM model – while its performance according to AIC is comparable to the standard linear regression model, its BIC is similar to that of the SLX.

Table 3.6

Model Comparison: Akaike and Bayesian Information Criteria, Log-Likelihood

Model	AIC	BIC	LogLik
NB	1233	1260	-608.7
Poisson	1527	1550	-756.4
SEM	1726	1756	-853.8
SLM	1726	1756	-853.8
SDM	1731	1781	-850.5
Linear Regression	1732	1758	-857.9
SLX	1739	1785	-855.3

4. Discussion

An examination of criminal records over the past seven years reveals that the level of drug-related crime in Czechia has remained relatively stable. During the period of the COVID-19 pandemic, we observe nothing but a marginal decline in drug offenses. We are of the opinion that this may be attributed to recent developments in the drug trade, which have been characterized by a shift toward online drug-dealing, enabling distribution networks to continue operating despite the government-imposed travel restrictions (Barratt & Aldridge, 2020; European Monitoring Centre for Drugs and Drug Addiction & Europol, 2020; Mravčík et al., 2021). Several factors may explain the slight decrease of detected criminality. One potential explanation is the disruption in the supply of precursors due to stricter border controls, which could have hampered the production capabilities of certain organized criminal groups. However, we cannot substantiate this claim with further evidence, as various studies did not report any changes in the price, purity or availability of drugs in Czechia during this period (European Monitoring Centre for Drugs and Drug Addiction & Europol, 2020; Mravčík et al., 2021). Furthermore, given that drug-related offenses are commonly classified as victimless crimes, their detection tends to be closely correlated with the intensity of law enforcement activities. Consequently, another contributing factor to the observed decline could have been the reduced operational capacity of the police, whose resources were primarily allocated to enforce pandemic-related public health measures during the peak of the crisis.

Spatial analysis revealed that the highest concentrations of drug activities are in territories along the national borders with Germany and Poland, supporting the hypothesis of the transnational nature of drug trade. The appeal of these regions to drug dealers is likely due to a combination of easier access to chemical precursors thanks to the proximity to Polish borders, as well as the opportunity to export drugs to Germany, and the increased prevalence of local addicts. Furthermore, these regions have long been characterized by lower living standards, resulting from relatively low income levels and persistently high long-term unemployment rates.

The results of the regression analysis presented herein are based on the estimates obtained from the negative binomial model. A one percentage point increase in the share of young men between the ages of 15 and 34 in population is associated with a 16.55% increase in the expected crime rate, while a one percentage point change in the divorce rate leads to a 21.48% increase. Similarly, a one percentage point change in the share of individuals with low education in population is accompanied by a 5.807% increase in the expected crime rate. Although a 1% change in the unemployment rate is expected to raise the crime rate by 7.228%, this effect was found to be insignificant at the 5% level. A 1% change in both the share of foreigners and the number of living allowances per 1 000 inhabitants corresponds to a 11.70% and 21.06% decrease in the crime rate, respectively. Both of these effects have, however, shown to be statistically insignificant. To conclude, these findings align with the theories outlined in Chapter 1 and point out that areas with weakened social ties, low education

attainment, and insufficient financial security of inhabitants are likely to exhibit elevated crime rates.

Conclusion

The aim of this thesis was to provide a broad analysis of drug-related criminality in the context of spatial econometrics.

The first two chapters were dedicated to establishing a theoretical foundation. In the first section, we offered an interdisciplinary literature review on crime in a socioeconomic context. It sought to deliver the fundamental idea that conceptualizes the importance of studying the prevalence of criminal behavior not only from the perspective of criminology, but also within the broader context of other social sciences, including economics. The sociological theories outlined in this section helped in understanding the underlying processes that might contribute to higher crime rates and served as the basis for selecting the explanatory variables used in the practical part of the thesis. The second section described the statistical and econometric tools commonly used in the analysis of data containing a spatial component.

The empirical research was based on data from the open database of the Czech Police, which provides georeferenced records of criminal offenses. Spatial analysis confirmed the positive autocorrelation of crime rate in Czechia and detected three clusters of administrative units (AD MEP) that exhibit high crime rates, all of which belonged to the northern or north-eastern regions of the Czech Republic. Building upon the findings, regression analysis was performed to explain the variation in crime across the studied territory using six preselected socioeconomic factors. In addition to the classical linear regression model, several specifications of spatial models were considered, and the analysis was extended to include models for count data, which—based on the evaluation using information criteria—proved to be most suitable. This was expected from the start, given the nature of the regressand. Among all the alternatives considered, the negative binomial model was selected as the most appropriate. According to the results, among the most influential predictors was the share of divorced individuals, which was found to have a significant positive effect on the crime rate. In line with theoretical expectations, the share of young men aged 15 to 34 years also showed a significant positive effect, linking this demographic group to increased participation in illegal activities. Similarly, a higher proportion of people with low education was significantly positively associated with crime. In contrast, the positive impact of the unemployment rate was not statistically significant at the 5% level. Negative, however insignificant, effects were reported for the remaining variables, the share of foreigners and the relative number of allowances for living.

Nevertheless, given the social nature of the data and the overall complexity of criminal behavior, many relevant factors remained unaccounted for, which contributed to the limited explanatory power of the models herein presented. This thesis faced several other limitations that should be acknowledged. The first stems from the fact that most drug-related crimes are victimless and their detection relies heavily on the proactive approach taken by police forces. As a result, these types of offenses are especially prone to underreporting, which undermines the reliability of the data source. Furthermore, we faced the obstacle of socioeconomic data

availability at the level of detail required for the statistical analysis.

Future research could benefit from exploring the impact of police activities on crime prevention by examining whether a higher probability of conviction leads to a notable reduction in unlawful behavior. Additionally, moving beyond a cross-sectional approach and adopting panel data methods would allow for the study of temporal trends and enable researchers to investigate how crime evolves over time in response to social changes and policy interventions.

In summary, this thesis demonstrates that socioeconomic indicators are statistically associated with regional crime rates and provides a methodological basis for further investigation into the geography of crime.

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